

EPIGENETIC CLOCK REASONING BENCH

Technical Documentation

Testing LLM Reasoning About Biological Aging

MESA Agent-Based Simulation · GEO-Calibrated Drift Rates · Algorithmic Ground Truth

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Repository: github.com/varunnair1234/CaltechLHacks_TeamSquad

1. Executive Summary

Large language models are increasingly used in longevity research to summarize literature, interpret biomarker data, and assist in clinical decision support. However, no rigorous benchmark exists to measure whether these models genuinely understand epigenetic clock biology — a core pillar of modern aging science. Models may produce fluent, confident-sounding responses while fundamentally misinterpreting clock values, conflating different clock types, or missing critical discordance patterns that carry clinical significance.

We present the Epigenetic Clock Reasoning Bench (ECRB) — a benchmark suite that generates biologically grounded clinical scenarios with algorithmically derived ground truth labels, then evaluates LLM responses across four categories of epigenetic clock reasoning. The benchmark uses MESA agent-based simulation calibrated against real GEO methylation data to generate realistic tissue aging scenarios, which are converted into natural language clinical vignettes by Claude API, then evaluated against ground truth labels that are computed entirely from simulation parameters without human annotation.

Our evaluation of two models — Claude Sonnet 4.6 and BioLLM (Longevity-Tuned) — across 200 scenarios reveals significant performance differences by task type, including a surprising finding: BioLLM outperforms Claude on intervention reasoning tasks (45.0% vs 40.0%), suggesting domain-specific training advantages for longitudinal clinical trajectory analysis despite Claude's superior overall performance (77.2% vs 55.4%).

Key Finding: LLMs struggle most with confounder awareness (Type D) and intervention reasoning (Type B), revealing critical gaps in biological reasoning that general-purpose language training does not address. These gaps have direct implications for AI-assisted longevity research.

Metric	Value
Total Scenarios Generated	200
Task Categories	4 (Clock Interpretation, Intervention Reasoning, Multi-Tissue, Confounders)
Models Evaluated	2 (Claude Sonnet 4.6, BioLLM Longevity-Tuned)
Training Data Source	GSE40279 (656 samples), GSE51057 (329 samples)
Ground Truth Method	Algorithmic (zero hand annotation)
Claude Sonnet 4.6 Overall Accuracy	77.2%
BioLLM Overall Accuracy	55.4%
Claude Parse Failures	0 / 200
BioLLM Parse Failures	16 / 200

2. Scientific Background

2.1 Epigenetic Clocks

Epigenetic clocks are machine learning models trained to predict biological age from DNA methylation patterns. Unlike chronological age, biological age reflects the functional state of tissues and correlates more strongly with health outcomes, disease risk, and mortality. Methylation marks — chemical modifications on cytosine bases at CpG dinucleotides — accumulate in predictable patterns across the genome as cells age, making them reliable molecular timestamps.

Three clocks form the primary evaluation targets of this benchmark:

Clock	Year	What It Measures	Primary Use Case
Horvath	2013	Pan-tissue biological age from 353 CpG sites	General aging, tissue comparisons, research baseline
GrimAge	2019	Mortality risk via plasma protein proxies	Lifespan prediction, cardiovascular disease risk
DunedinPACE	2022	Speed of aging (population mean = 1.0)	Intervention response measurement, rate tracking

Each clock captures a different dimension of biological aging, which creates clinically meaningful discordance patterns. A patient may show a favorable Horvath biological age while simultaneously exhibiting elevated GrimAge — suggesting that while their epigenome broadly reflects healthy aging, their specific mortality-risk protein expression pattern indicates elevated danger. These discordance patterns require nuanced biological reasoning to interpret correctly, making them ideal test cases for LLM evaluation.

2.2 The Hallmarks of Aging Relevance

The simulation underlying our benchmark draws from the López-Otín hallmarks of aging framework (2013, updated 2023). Specifically, our cell-level simulation models three interconnected hallmarks that drive methylation clock changes:

- Cellular senescence: cells that stop dividing but remain metabolically active, secreting the Senescence-Associated Secretory Phenotype (SASP) which damages neighboring cells and promotes systemic inflammation.
- Mitochondrial dysfunction: modeled through reactive oxygen species (ROS) accumulation that increases cellular damage counters.
- Epigenetic alterations: modeled through CpG methylation drift calibrated against real population data, producing biologically realistic Horvath clock trajectories.

The SASP signaling component — where senescent cells spray inflammatory signals to neighbors within a spatial radius — is specifically why we use agent-based modeling rather than differential equations. This local, stochastic, spatially-dependent cascade is not expressible as a simple ODE but emerges naturally from ABM cell-cell interaction rules.

2.3 Why LLM Benchmarking Matters for Longevity Research

The longevity research community faces a literature explosion: thousands of papers on senolytics, caloric restriction, mTOR inhibition, NAD⁺ precursors, rapamycin, and dozens of other interventions, each reporting clock measurements across different tissue types, age groups, and follow-up periods. AI tools are increasingly used to parse this literature and assist in patient-facing applications.

If these models cannot correctly interpret multi-clock profiles, identify technical confounders, or evaluate intervention trajectories, their deployment in research or clinical contexts carries real risk. You cannot improve what you cannot measure — ECRB provides the measurement infrastructure.

3. System Architecture

3.1 Three-Stage Pipeline

The ECRB pipeline has three stages that transform real biological data into evaluated benchmark results:

Stage	Input	Process	Output
1. Simulate	GEO drift rates, literature constants	MESA cell agents age over 200 timesteps	Tissue-level clock values per run
2. Generate	Clock values from simulation	Claude API generates NL clinical vignettes	benchmark.json (200 scenarios)
3. Evaluate	benchmark.json + model API access	LLMs answer, scorer grades responses	leaderboard.csv + details.json

3.2 Stage 1: MESA Cell Simulation

We use MESA 3.0, a Python agent-based modeling framework, to simulate tissue aging at the cellular level. Each simulation run creates a 30×30 grid (900 cells) representing a tissue patch, with each cell tracking individual biological state:

```
class CellAgent(mesa.Agent):
    cpg_methylation: dict[str, float] # 5 representative CpG sites
    damage_counter: float # ROS-driven damage accumulation
    telomere_length: float # Shortens each division (starts: 1.0)
    senescence_state: str # normal | stressed | senescent | dead
    sasp_level: float # 1.0 if senescent, 0.0 otherwise
    age_months: int # Simulation time counter
```

Each timestep represents one month of biological time. Simulations run 60-200 steps (5-16 years). The senescence cascade works as follows:

1. Cells accumulate damage from stochastic ROS events and SASP exposure from senescent neighbors within radius 2.
2. When telomere length falls below 0.15 (crisis threshold from Blackburn et al.), cells transition to stressed state.
3. Stressed cells with damage > 2.0 become senescent and begin secreting SASP.
4. Heavily damaged senescent cells (damage > 5.0) transition to dead, removing them from the tissue.
5. Tissue-level clock values are computed by aggregating live cell states.

3.2.1 Clock Proxy Computation

Three clock proxies are computed from the tissue state at each timestep:

```
# Horvath proxy: bulk methylation mean across live cells
horvath = mean([cell.mean_cpg_methylation for cell in live_cells]) * 100

# GrimAge proxy: combines senescent fraction and total SASP
grimage = senescent_fraction * 50 + total_sasp * 0.1

# DunedinPACE proxy: damage-rate and senescence-driven pace signal
dunedin = 1.0 + mean_damage * 0.1 + senescent_fraction * 0.5
```

3.2.2 Intervention Simulation

Type B benchmark scenarios require pre- and post-intervention clock values. Interventions are simulated by monkey-patching the ROS damage rate parameter during the post-intervention phase:

Intervention	Parameter Change	Biological Rationale
Caloric restriction	ROS rate $\times$ 0.70	CR reduces mitochondrial ROS production by 30% in model organisms
Senolytic therapy (D+Q)	Senescent fraction – 30% at intervention start	Dasatinib+Quercetin clears senescent cells directly
Aerobic exercise	ROS rate $\times$ 0.80, telomere rate $\times$ 0.85	Exercise upregulates antioxidant defenses and telomerase
Rapamycin	ROS rate $\times$ 0.75	mTOR inhibition reduces cellular stress and damage accumulation
NAD+ supplementation	ROS rate $\times$ 0.85	NAD+ restoration improves mitochondrial efficiency

3.3 Calibration Against Real GEO Data

A critical differentiator of ECRB is that simulation drift rates are not arbitrary — they are calibrated against real methylation data from NCBI GEO. This transforms the simulation from a toy model into a biologically grounded instrument.

Dataset	Samples	Age Range	Tissue	CpGs Used	Primary Use
GSE40279	656	19-101 years	Whole blood	102,307 ($r^2 > 0.02$)	Primary clock drift calibration
GSE51057	329	35-70 years	Multi-tissue	105,972 ($r^2 > 0.02$)	Tissue-specific drift ratios

The calibration pipeline regresses each CpG's beta value against chronological age across all samples, extracting the slope (drift rate per year) and R^2 value. Only CpGs with $R^2 > 0.02$ are retained, representing sites where age explains at least 2% of methylation variance. Five representative CpGs are selected as the simulation's primary methylation state, anchored by cg16867657 ($R^2 = 0.737$ in GSE40279 — the strongest age-correlated site in the Horvath 353 set):

CpG Site	R^2 (GSE40279)	Drift / Year	Drift / Month	Direction	Replicates in GSE51057
cg16867657	0.737	+0.00464	+0.000387	Increases	Yes
cg06639320	0.558	+0.00281	+0.000234	Increases	No
cg24724428	0.554	+0.00398	+0.000332	Increases	Yes
cg22454769	0.554	+0.00406	+0.000338	Increases	No
cg10501210	0.527	-0.00790	-0.000658	Decreases	No

Methodological Note: cg10501210 exhibits negative drift (methylation decreases with age), which is biologically real. Including both positive and negative drifting CpGs in the simulation produces more realistic bulk methylation trajectories than using only increasing sites.

Tissue-specific drift rates for Type C (multi-tissue discordance) scenarios are derived from the ratio of GSE51057 drift rates to GSE40279 rates for the three cross-validated CpGs. This ratio represents how much faster non-blood tissues age epigenetically relative to blood, grounding the tissue discordance scenarios in real cross-tissue methylation data.

3.4 Stage 2: Scenario Generation

Each simulation run produces one benchmark scenario through a two-step generation process:

- Simulation output (clock values, senescent fraction, tissue type, steps) is passed to the Claude Haiku API with a structured prompt requesting a clinical vignette in JSON format.
- Simultaneously, ground truth labels are computed algorithmically from the simulation parameters without any AI involvement.

The separation of natural language generation (AI) and ground truth labeling (algorithmic) is fundamental to the benchmark's validity. Ground truth labels are derived from published clinical thresholds:

```
def derive_ground_truth(sim_output, patient_age):
    horvath = sim_output['clocks']['horvath']
    grimage = sim_output['clocks']['grimage_proxy']
    dunedin = sim_output['clocks']['dunedinpace_proxy']
    sen_frac = sim_output['senescent_fraction']

    return {
        'accelerated_aging': horvath > patient_age + 5,
```

```
'fast_pacer':          dunedin > 1.1,
'high_mortality_risk': grimage > patient_age + 7,
'clock_discordance':   abs(horvath - grimage) > 5,
'high_senescence':     sen_frac > 0.12,
'intervention_effective': post_dunedin < pre_dunedin * 0.95 # Type B only
}
```

This approach enables infinite scalable scenario generation — each new simulation run automatically produces a fully labeled scenario. The benchmark can grow to thousands of scenarios without any manual annotation effort.

3.4.1 Task Type Distribution

Task Type	Count	What It Tests	Key Biological Reasoning Required
A: Clock Interpretation	80 (40%)	Multi-clock profile understanding	Horvath vs GrimAge vs DunedinPACE semantics, discordance patterns
B: Intervention Reasoning	50 (25%)	Pre/post intervention clock changes	Longitudinal trajectory analysis, intervention mechanism knowledge
C: Multi-Tissue Discordance	40 (20%)	Cross-tissue methylation differences	Tissue-specific drift rates, buccal vs blood vs brain clock behavior
D: Confounders & Artifacts	30 (15%)	Technical artifact identification	Cell composition shifts, medication effects, sample quality issues

4. Evaluation Methodology

4.1 Evaluation Harness

The evaluation harness sends each benchmark scenario's question to each model being evaluated and parses the structured JSON response. All models are evaluated under identical conditions with the same system prompt:

```
SYSTEM_PROMPT = '''
You are an expert in epigenetic aging biology and longevity research.
You will receive clinical scenarios involving epigenetic clock measurements.
Analyze the clock values and answer precisely.

Respond ONLY in this exact JSON format with no markdown:
{
  "accelerated_aging": true or false,
  "fast_pacer": true or false,
  "high_mortality_risk": true or false,
  "clock_discordance": true or false,
  "reasoning": "2-3 sentence explanation",
  "confidence": "high or medium or low"
}
'''
```

A 0.5-second delay between API calls prevents rate limiting. JSON parse failures are logged and counted separately — a model that cannot produce valid structured output on a given scenario receives zero points for that scenario.

4.2 Scoring Framework

Each scenario is scored on five dimensions for a maximum of 100 points:

Dimension	Points	Evaluation Method
Accelerated aging classification	25	Binary: predicted == ground_truth
Fast pacer identification	20	Binary: predicted == ground_truth
High mortality risk classification	20	Binary: predicted == ground_truth
Clock discordance detection	20	Binary: predicted == ground_truth
Reasoning quality	15	Keyword scoring: +5 per correct clock name cited (max 15)

A response is considered correct if it achieves a total score ≥ 60 points. This threshold reflects that a partially correct answer (getting 3 of 4 binary dimensions right) should count as a passing response, while getting all binary dimensions wrong but providing verbose reasoning should not.

The reasoning quality dimension specifically rewards models that cite clock names (Horvath, GrimAge, DunedinPACE) in their explanation, incentivizing responses that demonstrate mechanism awareness rather than black-box pattern matching.

4.3 Sensitivity Analysis with SALib

The scorer uses SALib (Sensitivity Analysis Library) to compute which scoring dimensions most strongly drive overall model performance. Sobol sensitivity indices reveal the variance contribution of each dimension to the total score variance. This allows us to answer: which aspects of epigenetic clock reasoning most differentiate models?

5. Results

5.1 Overall Leaderboard

Rank	Model	Overall Accuracy	Mean Score	Type A	Type B	Type C	Type D	Parse Fails
1	Claude Sonnet 4.6	77.2%	76.0	96.1%	40.0%	88.1%	66.2%	0 / 200
2	BioLLM (Longevity-Tuned)	55.4%	54.8	63.9%	45.0%	63.9%	34.6%	16 / 200

5.2 Key Findings

Finding 1: Claude Dominates Clock Interpretation (Type A)

Claude Sonnet achieves 96.1% accuracy on Type A tasks — straightforward interpretation of single-snapshot clock profiles. This represents near-ceiling performance and suggests that basic epigenetic clock knowledge is well-represented in Claude's training data. BioLLM achieves 63.9%, indicating meaningful knowledge but more frequent misclassification of clock semantics.

Finding 2: BioLLM Leads on Intervention Reasoning (Type B)

Surprising Result: BioLLM outperforms Claude Sonnet on Type B intervention reasoning (45.0% vs 40.0%). This is the only task category where the domain-specific model leads. This suggests that training on biomedical literature may confer specific advantages for reasoning about longitudinal clinical trajectories and intervention response dynamics, even at lower overall performance.

The absolute performance on Type B is low for both models (below 50%), suggesting that intervention trajectory reasoning is genuinely difficult. Models must understand how different clocks respond at different timescales — DunedinPACE responds rapidly to lifestyle interventions while GrimAge may take years to change. This temporal reasoning requirement exceeds what static clock interpretation tasks require.

Finding 3: Both Models Struggle with Confounders (Type D)

Claude achieves 66.2% and BioLLM 34.6% on confounder scenarios. The gap between Claude's Type A (96.1%) and Type D (66.2%) performance — 30 percentage points — reveals that even strong models fail to account for technical artifacts when interpreting clock values. Confounders tested include acute infection (elevated WBC), medication effects (metformin, antidepressants), cell composition shifts (chemotherapy), and sample quality issues.

This finding has direct practical implications: AI tools deployed to interpret patient clock results may confidently report accelerated aging when the signal is actually driven by an acute inflammatory

state or medication effect. Without confounder awareness, these tools could cause patient harm through unnecessary alarm.

Finding 4: Multi-Tissue Discordance Reveals Tissue Knowledge Gap (Type C)

Claude achieves 88.1% on Type C, suggesting strong but imperfect understanding of tissue-specific methylation differences. BioLLM's 63.9% performance indicates more frequent failures to account for the fact that brain tissue ages epigenetically approximately 2x faster than blood, or that buccal samples show different patterns than blood draws from the same individual.

Finding 5: Parse Reliability Strongly Differentiates Models

Claude produced zero JSON parse failures across 200 scenarios. BioLLM produced 16 parse failures (8%), meaning it failed to generate valid structured output on those scenarios, scoring zero points regardless of biological reasoning quality. In a production research context, an 8% failure-to-respond rate would be clinically unacceptable. Parse reliability is therefore a critical metric alongside accuracy.

6. Technical Stack

6.1 Backend Dependencies

Package	Version	Purpose
mesa	>=3.0,<4.0	Agent-based cell simulation framework
numpy	>=1.24	Numerical operations, random processes, drift rate arrays
pandas	>=2.0	Data collection, GEO dataset processing, scenario dataframes
scipy	>=1.10	Statistical distributions for parameter sampling
anthropic	latest	Claude API for scenario generation and model evaluation
SALib	latest	Sensitivity analysis for scoring dimension importance
GEOparse	>=2.0	GEO dataset download and SOFT file parsing
tqdm	latest	Progress bars for benchmark generation and evaluation runs
pydantic	>=2.0	Schema validation for benchmark JSON entries
pytest	>=7.0	Unit tests for simulation correctness and biology validation

6.2 Frontend Stack

The visualization dashboard is built with React and Tailwind CSS, using deck.gl for the GPU-accelerated tissue simulation visualization and Recharts for leaderboard charts and score breakdowns.

Component	Technology	Purpose
Framework	React + TypeScript	Component-based UI with type safety
Styling	Tailwind CSS	Dark theme utility classes
3D Visualization	deck.gl (GridCellLayer)	30x30 animated tissue simulation grid
Charts	Recharts	Grouped bar charts, radar charts, score breakdowns
Build	Vite / Lovable	Fast development and deployment
Hosting	Lovable deployment	One-click deployment from Lovable platform

6.3 Repository Structure

```
CaltechLHacks_TeamSquad/
```

```
|— simulation/
|   |— cell_agent.py          # MESA CellAgent with methylation/damage/telomere
state
|   |— tissue_model.py       # TissueModel: 30x30 grid, clock aggregation
|   |— clocks.py             # Horvath/GrimAge/DunedinPACE proxy formulas
|   |— params.py             # Literature-grounded simulation parameters
|   |— horvath_coefficients.csv
|— benchmark/
|   |— scenario_generator.py  # Simulation -> Claude API -> clinical vignette
|   |— ground_truth.py       # Algorithmic label derivation from sim params
|   |— benchmark.json        # 200 generated scenarios (306 KB)
|— eval/
|   |— clients.py            # Claude + BioLLM API client wrappers
|   |— scorer.py             # SALib-integrated scoring + leaderboard
|   |— run_eval.py           # CLI evaluation orchestrator
|— eval_outputs/
|   |— leaderboard.csv       # Model comparison scores
|   |— details.json          # Per-scenario per-model results (503 KB)
|— data/
|   |— GSE40279/             # drift_rates.csv, metadata.csv, horvath_cpgs.csv
|   |— GSE51057/            # drift_rates.csv, metadata.csv, horvath_cpgs.csv
|— frontend/                 # React/deck.gl dashboard
|— pipeline.py               # Master orchestrator (--demo, --generate, --
evaluate)
|— requirements.txt
```

7. Literature & Parameter Sources

7.1 Simulation Parameters from Published Literature

Parameter	Value Used	Source
Senescent fraction, young tissue	2-5%	Campisi et al. 2013, Nature Reviews Molecular Cell Biology
Senescent fraction, aged tissue	10-15%	van Deursen 2014, Nature
Telomere crisis threshold	~1.5 kb (normalized: 0.15)	Blackburn, Epel & Lin, Science 2015
SASP spread radius	5-10 cell diameters	Acosta et al. 2013, Nature Cell Biology
Horvath clock error	±3.6 years	Horvath 2013, Genome Biology
DunedinPACE mean / std	1.0 / ~0.1	Belsky et al. 2022, eLife
Brain CpG drift vs blood	~2x faster	Multi-tissue clock papers, Horvath et al.
Caloric restriction ROS reduction	30%	Colman et al. 2009, Science
Exercise telomere effect	15% reduction in shortening rate	Werner et al. 2019, European Heart Journal

7.2 Epigenetic Clock References

- Horvath S. (2013). DNA methylation age of human tissues and cell types. *Genome Biology*, 14(10), R115.
- Lu AT, et al. (2019). DNA methylation GrimAge strongly predicts lifespan and healthspan. *Aging*, 11(2), 303-327.
- Belsky DW, et al. (2022). DunedinPACE, a DNA methylation biomarker of the pace of aging. *eLife*, 11, e73420.
- López-Otín C, et al. (2013). The hallmarks of aging. *Cell*, 153(6), 1194-1217.
- López-Otín C, et al. (2023). Hallmarks of aging: An expanding universe. *Cell*, 186(2), 243-278.
- Levine ME, et al. (2018). An epigenetic biomarker of aging for lifespan and healthspan (PhenoAge). *Aging*, 10(4), 573-591.

7.3 GEO Dataset Citations

- Hannum G, et al. (2013). Genome-wide methylation profiles reveal quantitative views of human aging rates. *Molecular Cell*, 49(2), 359-367. [GSE40279]
- Lehne B, et al. (2015). A coherent approach for analysis of the Illumina HumanMethylation450 BeadChip improves data quality and performance in epigenome-wide association studies. *Genome Biology*, 16(1), 37. [GSE55763]
- Horvath S, et al. (2012). Aging effects on DNA methylation modules in human brain and blood tissue. *Genome Biology*, 13(10), R97. [GSE51057]

8. Limitations and Future Work

8.1 Current Limitations

- Clock proxy formulas are approximations derived from simulation parameters, not actual methylation array computation. Production-grade evaluation would use real DNA methylation data processed through pyaging's validated clock implementations.
- The benchmark currently evaluates two models. Expanded evaluation against GPT-4o, Gemini 1.5 Pro, and other bio-specialized LLMs would provide a more comprehensive landscape of LLM epigenetic reasoning capabilities.
- Type B intervention scenarios use ROS parameter modification to simulate interventions, which is a simplified proxy for the complex molecular mechanisms of each intervention type.
- All scenarios are generated from simulated data. While calibrated against real GEO data, validation against real clinical cases from longitudinal cohorts (MESA, TwinsUK) would strengthen the benchmark's clinical relevance.
- The current ground truth labeling does not account for clinical context beyond the simulation parameters, meaning some borderline cases may be labeled differently than a human expert would label them.

8.2 Future Roadmap

- Clinical cohort validation: Apply benchmark scenarios derived from MESA (dbGaP phs000209) and TwinsUK longitudinal methylation data to validate simulation-derived scenarios against real clinical outcomes.
- Expanded model evaluation: Evaluate GPT-4o, Gemini 1.5 Pro, BioMedLM, and Insilico's fine-tuned LongevityLLM to build a comprehensive benchmark leaderboard.
- Agentic pipeline integration: Extend the benchmark to automatically generate scenarios from new longevity research publications using PubMed API and NLP extraction.
- LongevityBench contribution: Export benchmark in standardized format for community contribution to emerging LLM evaluation frameworks in longevity biology.
- Scale to 1,000+ scenarios: The algorithmic generation pipeline scales without cost beyond API expenses. Running 800 additional scenarios would provide stronger statistical power for model comparison.
- GrimAge v2 and DunedinPACE v2 tasks: Update clock proxies to reflect the latest clock versions and add task categories specifically testing knowledge of clock evolution and methodological improvements.

9. Reproducibility and Setup

9.1 Requirements

```
# Python 3.11+ required (3.14 requires --ignore-requires-python for some packages)

# Install dependencies
pip install -r requirements.txt

# Set API key
export ANTHROPIC_API_KEY='your-key-here' # Unix/Mac
$env:ANTHROPIC_API_KEY = 'your-key-here' # Windows PowerShell
```

9.2 Running the Pipeline

```
# Full pipeline: generate + evaluate
python pipeline.py --full

# Demo mode: 5 scenarios (fast, for testing)
python pipeline.py --demo

# Generate benchmark only
python pipeline.py --generate --scenarios 200

# Evaluate models on existing benchmark
python pipeline.py --evaluate

# Run tests
pytest tests/ -v
```

9.3 GEO Data Setup

GEO datasets are downloaded and processed by the `geo_scraper.py` script. The resulting CSV files (`drift_rates.csv`, `metadata.csv`, `horvath_cpGs.csv`) are committed to the repository in the `data/` folder. Raw SOFT files and beta matrices are excluded from version control due to size (>7 GB unprocessed).

```
# Download and process GEO data (one-time setup, ~30 min)
python geo_scraper.py

# Data will be saved to:
# data/GSE40279/drift_rates.csv (7.5 MB, 102k CpGs with  $r^2 > 0.02$ )
# data/GSE40279/metadata.csv (656 samples, age 19-101)
# data/GSE51057/drift_rates.csv (7.6 MB, 106k CpGs with  $r^2 > 0.02$ )
```

```
# data/GSE51057/metadata.csv      (329 samples, age 35-70)
```

Appendix A: Sample Benchmark Scenarios

Type A: Clock Interpretation

Scenario #0 | Task Type A | Patient Age: 70 | Tissue: Blood | Simulation: 187 steps Patient Context: A 70-year-old male with a 10-year history of type 2 diabetes and hypertension presents for comprehensive aging assessment after his primary care physician noted metabolic syndrome markers. Clock Values: Horvath 57.6 years | GrimAge 86.9 years | DunedinPACE 1.14 | Senescent Fraction 7.3% Question: This patient shows a striking discordance between epigenetic clocks: Horvath clock suggests favorable aging (13.4 years younger than chronological age) while GrimAge suggests accelerated aging (16.9 years older). The DunedinPACE of 1.140 indicates aging pace 14% faster than normal. How would you reconcile these conflicting signals, and which clock or combination of findings should most influence clinical recommendations regarding his diabetes and metabolic risk? Ground Truth: accelerated_aging=False | fast_pacer=True | high_mortality_risk=True | clock_discordance=True | overall_status=at_risk

Type D: Confounders and Artifacts

Scenario #170 | Task Type D | Patient Age: 51 | Tissue: Buccal | Simulation: 142 steps Confounder: Blood sample drawn during acute infection (elevated WBC) Patient Context: A 51-year-old patient with an acute infection (elevated WBC) underwent epigenetic clock testing via buccal sample collection during the acute phase of illness. Clock Values: Horvath 53.9 years | GrimAge 62.3 years | DunedinPACE 1.025 | Senescent Fraction 3.3% Question: Given that this patient's GrimAge estimate (62.3 years) shows an 11.3-year acceleration compared to chronological age while Horvath clock shows only 2.9 years acceleration, and the sample was collected during acute infection with elevated WBC, how should the clinician interpret the discordance between these clocks? Should the results be repeated after infection resolution, and which clock's results are more likely confounded by the acute inflammatory state? Ground Truth: accelerated_aging=False | fast_pacer=False | high_mortality_risk=True | clock_discordance=True | overall_status=at_risk

Appendix B: Glossary

Term	Definition
CpG site	Cytosine-guanine dinucleotide where DNA methylation occurs; methylation levels at these sites change predictably with age
Beta value	Methylation level at a CpG site, ranging from 0 (unmethylated) to 1 (fully methylated)

Term	Definition
Biological age	Estimated age based on biomarker patterns rather than chronological age; can diverge significantly from birth date
DunedinPACE	Pace of aging metric; values above 1.0 indicate faster-than-average aging, below 1.0 indicates slower
SASP	Senescence-Associated Secretory Phenotype; inflammatory molecules secreted by senescent cells that damage neighboring tissue
Inflammaging	Chronic, low-grade sterile inflammation associated with aging; driven in part by SASP accumulation
Hallmarks of aging	Cellular and molecular mechanisms underlying aging, as defined by López-Otín et al. 2013/2023
ABM	Agent-Based Model; simulation where individual agents (cells) follow rules and emergent behavior arises from interactions
ROS	Reactive Oxygen Species; byproducts of mitochondrial metabolism that accumulate with age and cause cellular damage
Clock discordance	When two epigenetic clocks give substantially different biological age estimates for the same sample; often clinically meaningful

Appendix C: Statistical Rigor Analysis

This appendix documents the class distribution, F1 scores, and statistical rigor analysis added to the benchmark evaluation after initial results were obtained. These metrics directly address the judging criteria for the AI Longevity Track at the Caltech Longevity Hackathon 2026.

C.1 Class Distribution

The following table shows the distribution of positive labels across all 200 benchmark scenarios. Class imbalance is a critical consideration for interpreting accuracy-based metrics.

Label	Positive Count	% Positive	Balance Assessment
accelerated_aging	68 / 200	34.0%	Reasonably balanced
fast_pacer	18 / 200	9.0%	Imbalanced — minority class
high_mortality_risk	168 / 200	84.0%	Imbalanced — majority class
clock_discordance	151 / 200	75.5%	Skewed positive
high_senescence	0 / 200	0.0%	Critical issue — threshold never reached

Critical finding: high_senescence is 0/200 positive across all scenarios. The senescent fraction threshold of 0.12 in derive_ground_truth() is never reached by the MESA simulation under default parameters. This label contributes 0 useful signal. Recommended fix for v2: lower threshold to 0.05 or derive it empirically from the simulation output distribution.

Imbalance note: fast_pacer (9% positive) and high_mortality_risk (84% positive) are heavily skewed. A naive always-false classifier achieves 91% accuracy on fast_pacer; a naive always-true classifier achieves 84% accuracy on high_mortality_risk. Raw accuracy is therefore misleading for these labels — F1 score is the appropriate metric.

C.2 F1 Scores

F1 score (harmonic mean of precision and recall) is reported for each label to account for class imbalance. high_senescence is included for completeness but should be interpreted as a benchmark construction issue, not a model capability signal.

Label	Claude Sonnet 4.6 F1	BioLLM F1	Delta (Claude - BioLLM)
accelerated_aging	0.942	0.551	+0.391
fast_pacer	0.947	0.627	+0.320
high_mortality_risk	0.910	0.717	+0.193
clock_discordance	0.902	0.805	+0.097
high_senescence (invalid)	0.000	0.000	0.000
Average (4 valid labels)	0.925	0.675	+0.250

C.3 Baseline Comparison

To contextualize model performance, two baselines are reported:

Baseline	Overall Accuracy	Interpretation
Random (50/50 binary guessing)	~50%	Lower bound — pure chance
Always-majority (predict most common class)	~60-70%	Naive baseline exploiting imbalance
BioLLM (domain-specific)	55.4%	Marginally above random overall
Claude Sonnet 4.6	77.2%	27 points above random baseline

Claude Sonnet 4.6 achieves 77.2% overall accuracy and 0.925 average F1 across 4 valid labels, representing a meaningful and robust performance above both random and naive baselines. BioLLM at 55.4% accuracy and 0.675 average F1 is above random but substantially below Claude on most labels, with the exception of intervention reasoning (Type B) where BioLLM leads.

C.4 Recommended v2 Improvements

- Lower high_senescence threshold from 0.12 to 0.05 to ensure positive labels appear in the benchmark
- Rebalance fast_pacer and high_mortality_risk by stratified sampling during scenario generation
- Report balanced accuracy and macro-averaged F1 as primary metrics in addition to raw accuracy
- Add a random baseline model to the leaderboard for interpretability

Appendix D: Claude API Interpretation Layer

The dashboard integrates Claude API to translate raw benchmark results into plain-language explanations accessible to non-technical users — longevity researchers, clinicians, and institutional stakeholders who need to understand what model performance means without parsing F1 scores or JSON output.

D.1 Motivation

Benchmark results in their raw form are opaque to non-specialists:

```
{
  "accelerated_aging": false,
  "fast_pacer": true,
  "high_mortality_risk": true,
  "clock_discordance": true,
  "total_score": 65
}
```

A longevity researcher seeing this output needs to know: what does clock_discordance mean clinically? Why is high_mortality_risk true when accelerated_aging is false? What should a practitioner do with this information? The Claude API interpretation layer answers these questions automatically.

D.2 Architecture

The interpretation pipeline follows three steps:

- Structured context assembly: benchmark scores, clock values, task type, patient age, and ground truth labels are assembled into a structured prompt
- Claude API call: the structured context is sent to claude-sonnet-4-6 with a longevity-tuned system prompt
- Plain-language output: Claude returns a 3-4 sentence explanation rendered directly on the dashboard scenario browser

D.2.1 System Prompt Design

The system prompt is carefully engineered to ensure explanations are grounded, accurate, and appropriately scoped:

SYSTEM_PROMPT = ""

You are a longevity health analyst interpreting epigenetic clock benchmark results for a research dashboard. Your audience includes clinicians, researchers, and non-technical stakeholders.

Guidelines:

- Explain what the clock values and scores mean in plain language
- Reference specific clock names (Horvath, GrimAge, DunedinPACE)
- Explain discordance patterns when present
- Suggest the most relevant clinical or research implication
- Do NOT provide medical diagnoses or treatment recommendations
- Always note this is a research tool, not a clinical device
- Keep explanation to 3-4 sentences maximum

"""

D.2.2 Example Input and Output

Input to Claude API (structured context from benchmark):

Patient: 70-year-old male
 Horvath clock: 57.6 years (12.4 years younger than chronological)
 GrimAge: 86.9 years (16.9 years older than chronological)
 DunedinPACE: 1.14 (aging 14% faster than average)
 Senescent fraction: 7.3%
 Model scored: fast_pacer=True, high_mortality_risk=True,
 clock_discordance=True, accelerated_aging=False
 Task type: A (clock interpretation)
 Model total score: 65/100

Claude API output displayed on dashboard:

This patient's results show a complex aging pattern. While their overall biological age appears normal on the Horvath clock, two signals stand out: GrimAge — which predicts mortality risk via plasma protein proxies — is significantly elevated at 16.9 years above chronological age. Their DunedinPACE of 1.14 means they are aging 14% faster than average. The discordance between clocks suggests metabolic and cardiovascular aging is outpacing cellular aging. Research tool only — not for clinical diagnosis.

D.3 Dashboard Integration

The interpretation panel appears alongside each scenario in the dashboard scenario browser. Users can:

- View the plain-language explanation for any of the 200 benchmark scenarios
- See which model scored the scenario and how the explanation relates to the score
- Compare explanations across Claude and BioLLM responses for the same scenario
- Understand why a model answer was marked correct or incorrect in accessible terms

Design decision: Explanations are generated from the ground truth scenario data, not from the model's raw response. This ensures the explanation is always accurate regardless of model performance — it explains what the correct answer should be, helping researchers understand the biological reasoning required for each task type.

D.4 Prompt Engineering Principles

Four principles guided the system prompt design:

- Grounded in data: explanations reference specific clock values and scores from the scenario, not generic statements about aging
- Mechanistically accurate: explanations use correct biological terminology including SASP, mortality risk proxies, and pace of aging rather than simplified analogies that sacrifice accuracy
- Scoped appropriately: the system prompt explicitly prevents diagnostic claims, ensuring compliance with wellness tool framing required for non-clinical deployment
- Consistent format: every explanation follows the same structure covering what the clocks show, what the pattern means, and what the implication is, enabling comparability across scenarios